# Cosmology

## Galaxies and Beyond

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University of Cambridge

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# Overview

What is Cosmology?

Our Place in the Cosmos

The Big Bang

Dark Matter

Dark Energy

$t = 0$

Modern Astronomy

# What is Cosmology?

**Cosmology is the study of the universe as a whole**

Age: How old is the universe?

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| Content: | What is Dark Matter and Dark Energy? |

# The Night Sky

## The Milky Way

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# Ground based telescopes

## The Coma Cluster

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# Space Telescopes

## Hubble Extreme Deep Field

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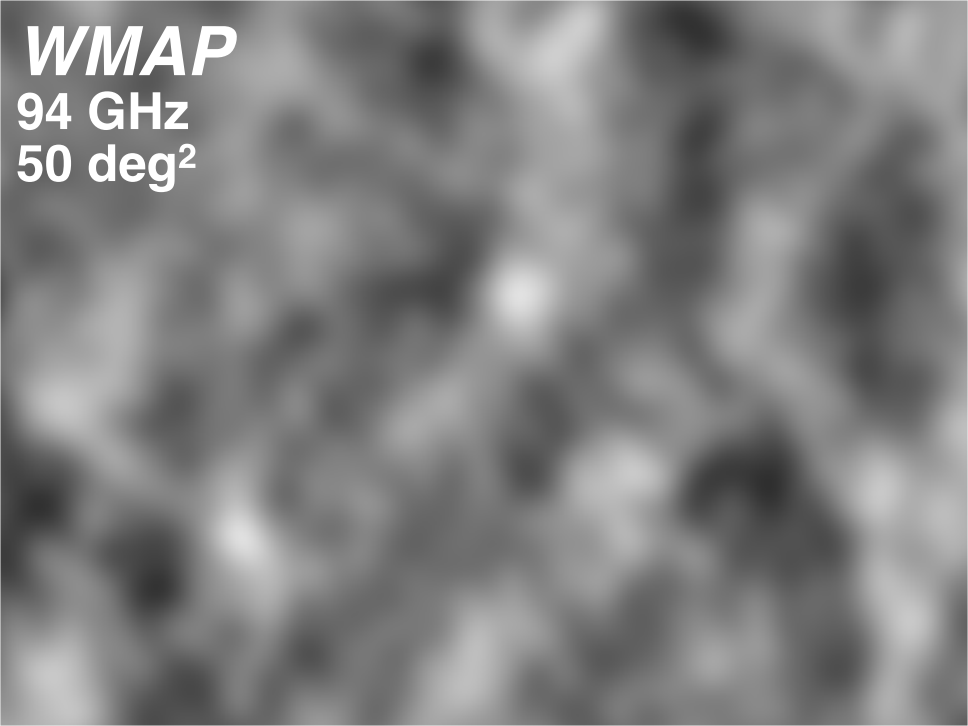
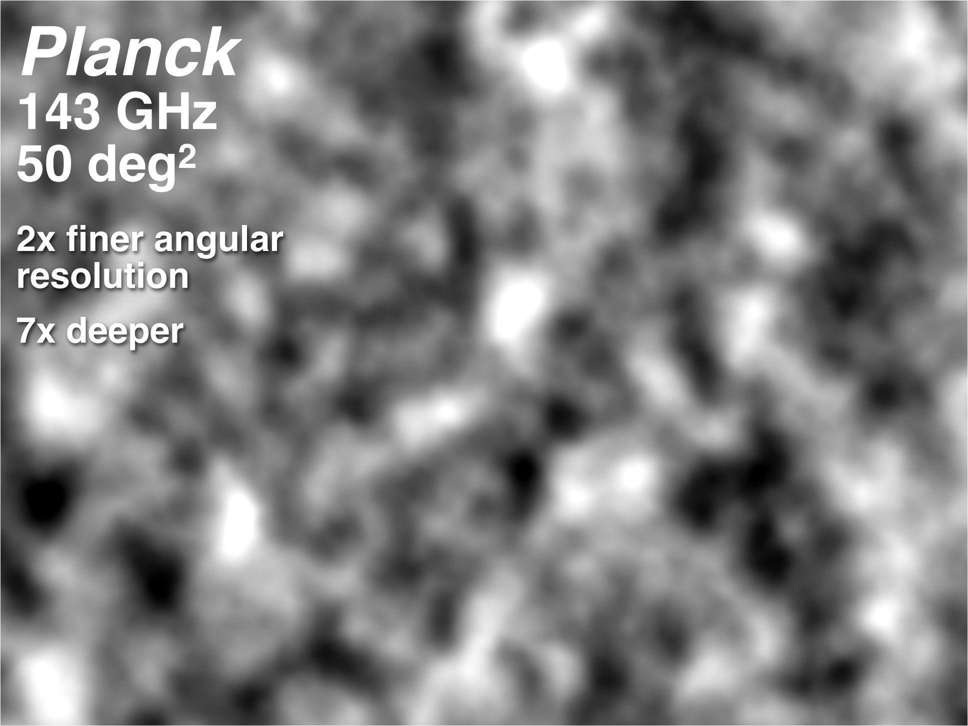
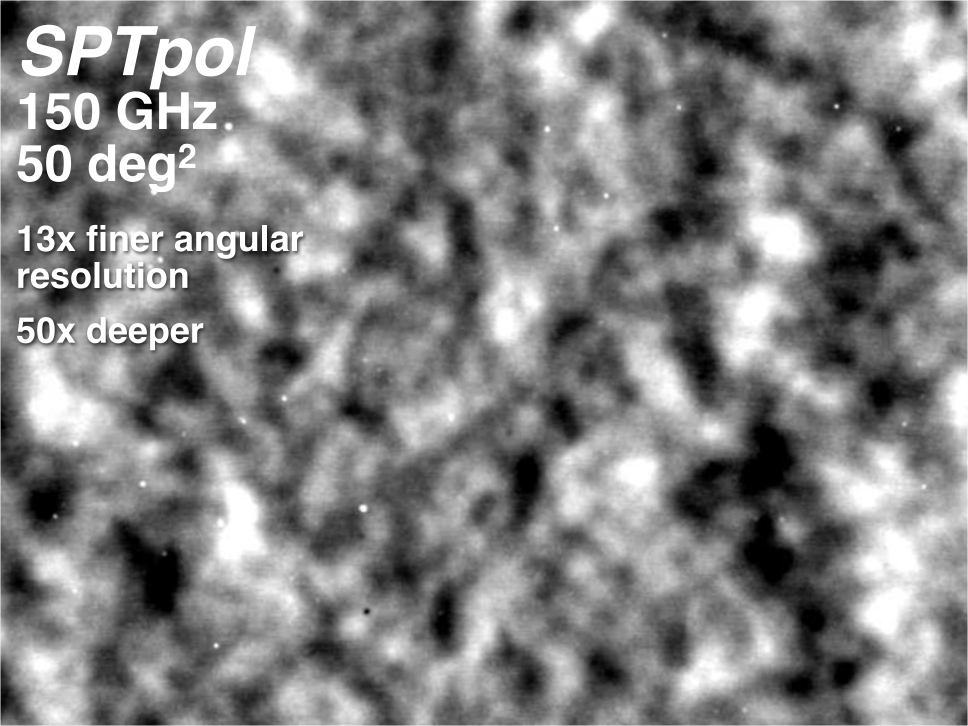
# The Dark Ages

# The CMB

## Cosmic Microwave Background


WMAP
94 GHz
50 deg²


*Planck*
143 GHz
50 deg²
2x finer angular resolution
7x deeper


SPTpol
150 GHz
50 deg²
13x finer angular resolution
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# What is the CMB?

- In order to understand what this CMB radiation is, we need to dispel some common misconceptions associated with "The Big Bang".

# The Big Bang

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## Galactic Redshift

As observed by Edwin Hubble in 1929, it appears that:

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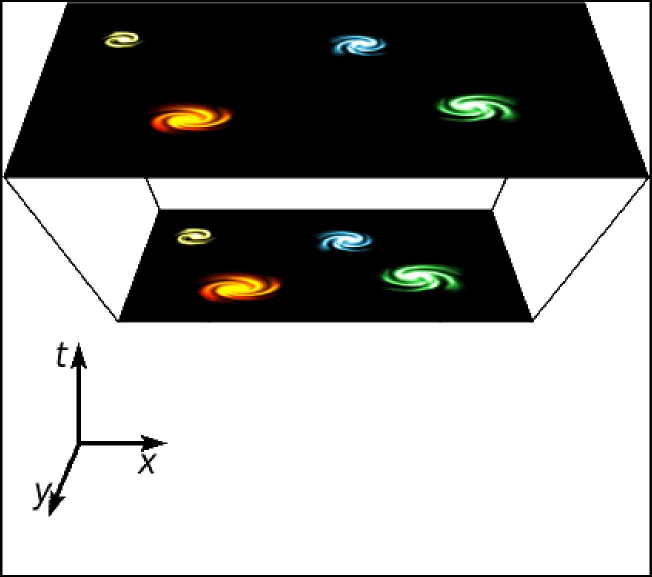
2. The Further away they are, the faster they are moving away
3. If we were to go to another galaxy, we would see exactly the same thing

# The Big Bang

**A stretching universe:**



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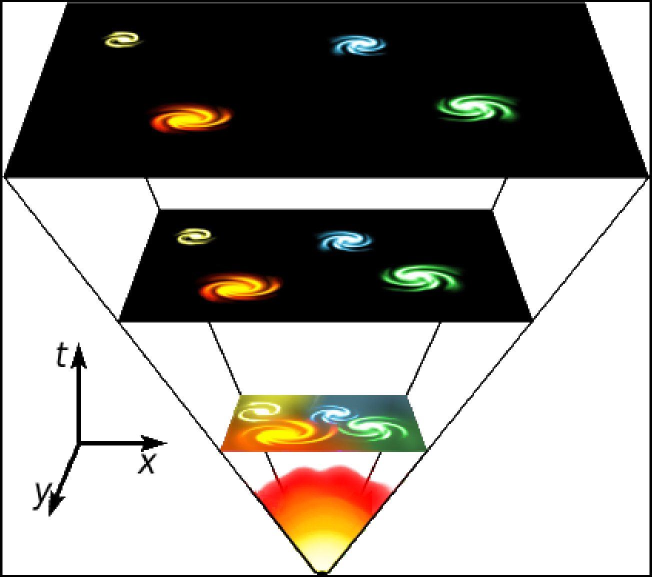

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## Common Misconceptions

The Big Bang does *NOT* look like this:

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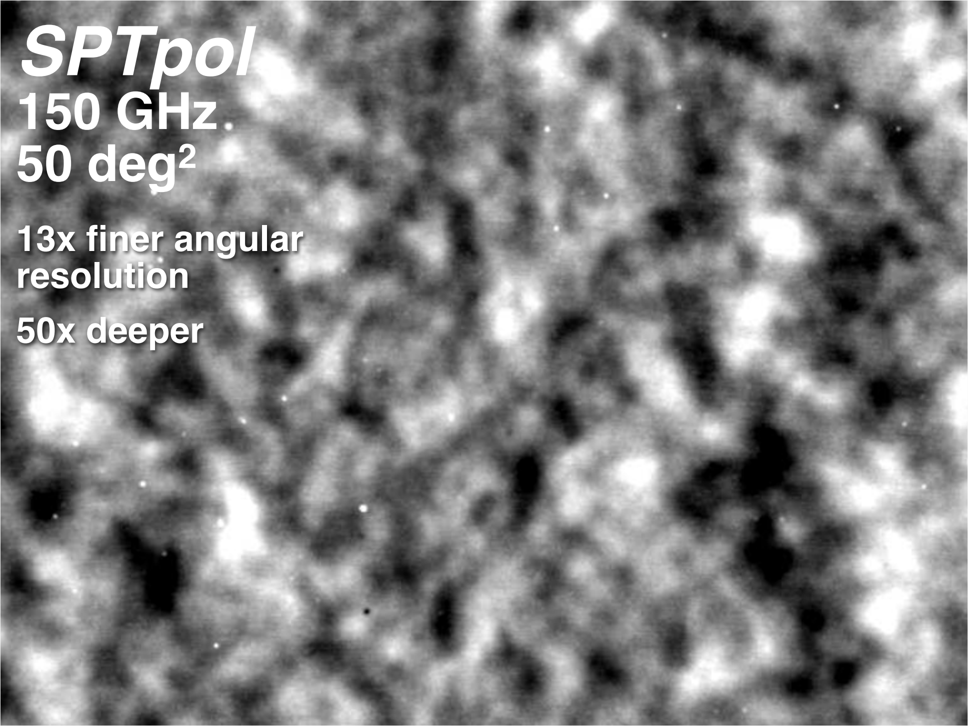
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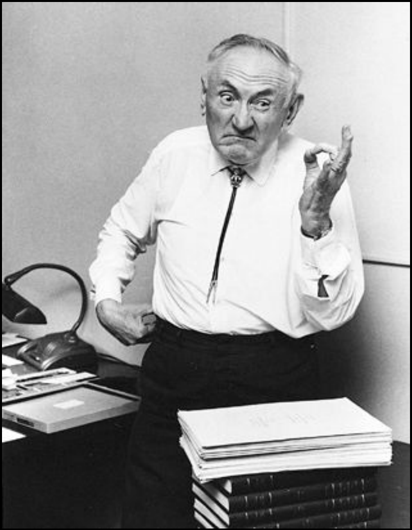

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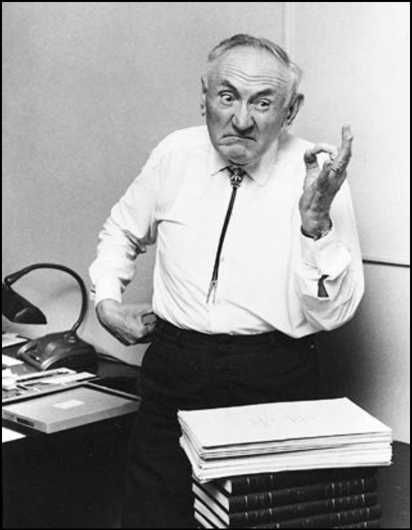

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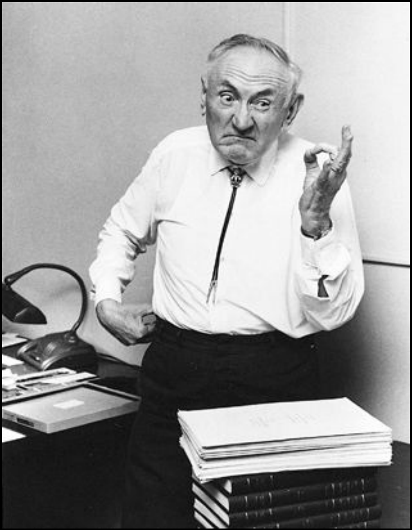

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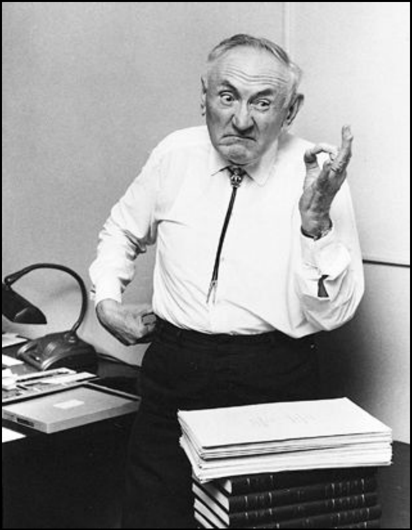

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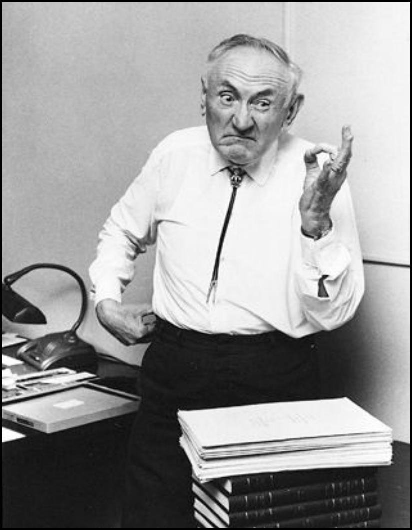
- Right now it's really just a name for a phenomenon we don't understand yet.



# Dark Energy

## The Problem

# The evolution of the universe

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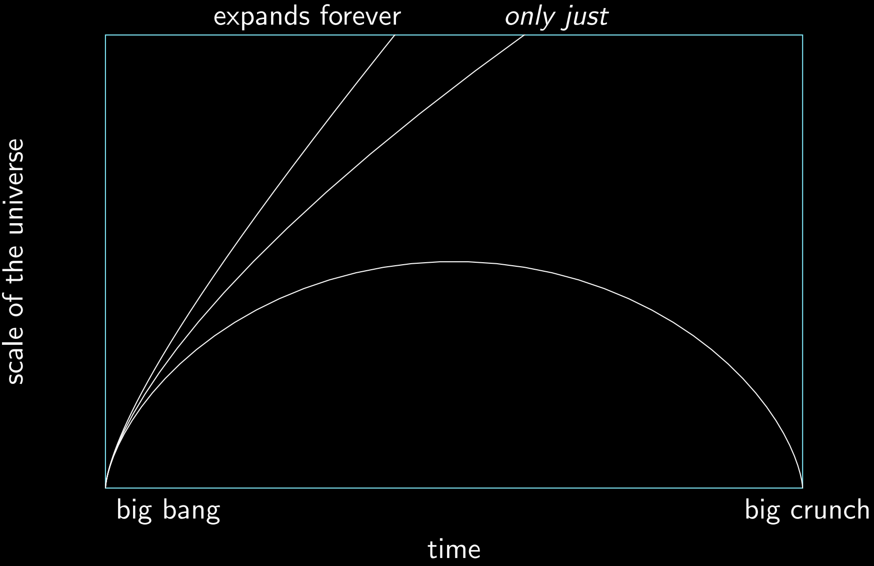

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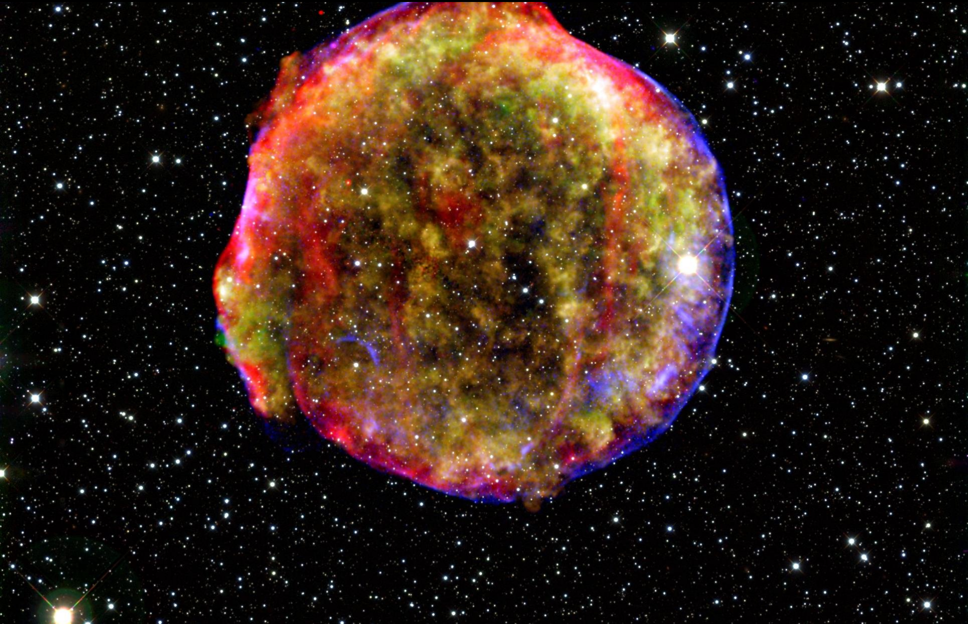

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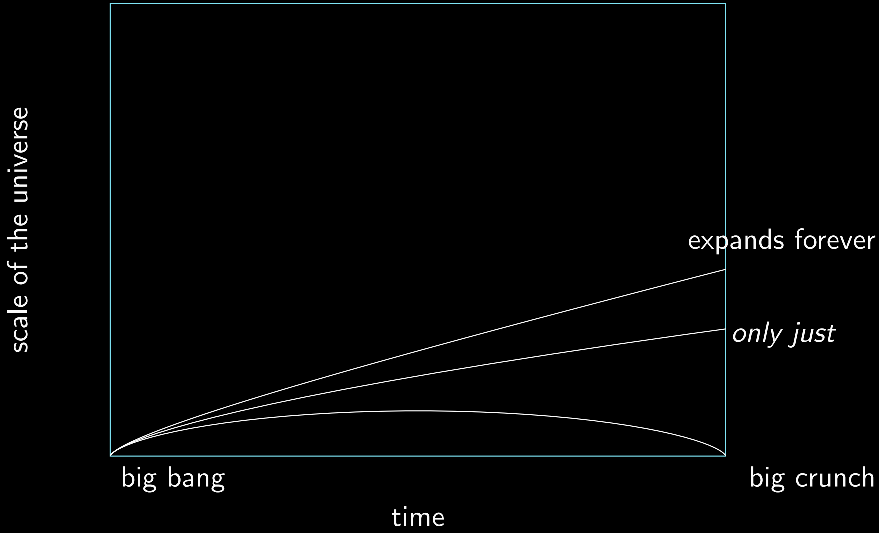
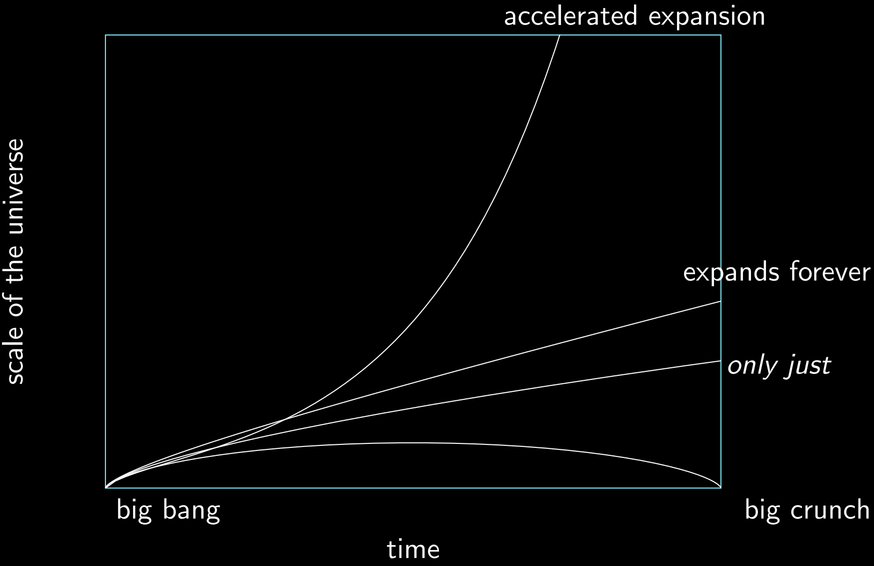


# The evolution of the universe

**None of the above:**



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- The Universe isn't slowing down in its expansion, but *speeding up!*
- One of the ways in which this could happen would be the presence of an energy permeating the universe, accelerating the expansion.
- We call this Dark Energy
- Like Dark Matter, it's really just a name for a phenomenon which we observe, and haven't been able to explain yet

# Dark Energy. . .

**. . . or Cosmological Constant?**

- Einstein's gravity can be repulsive at large scales.

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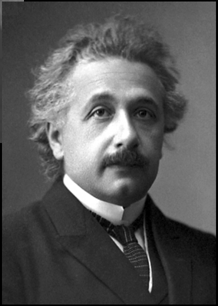

# Dark Energy...

## ...or Cosmological Constant?

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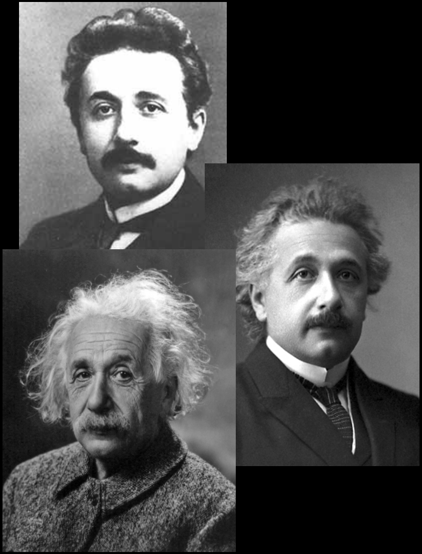
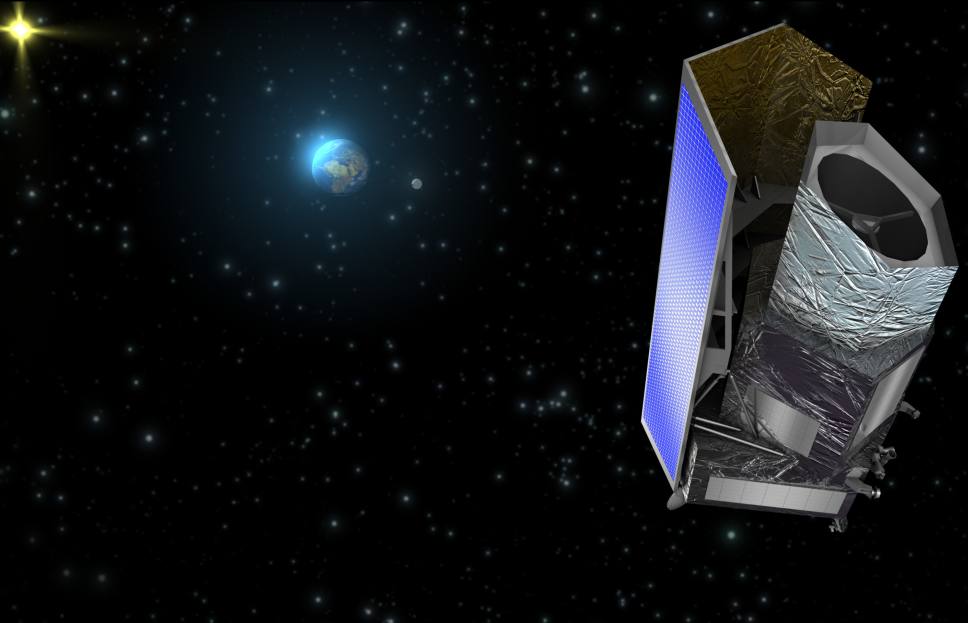
- This is known as a cosmological constant.
- In modern terms, we think of it as a type of energy.



# The Euclid Satellite

## Exploring dark energy



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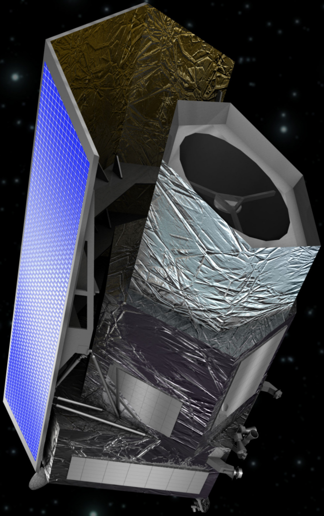

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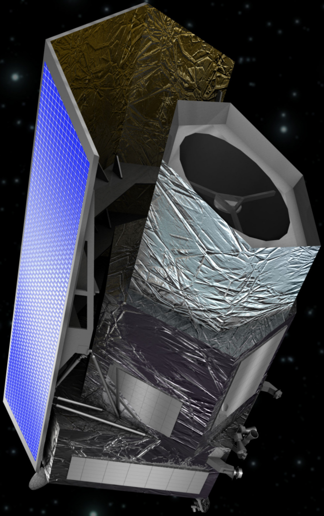

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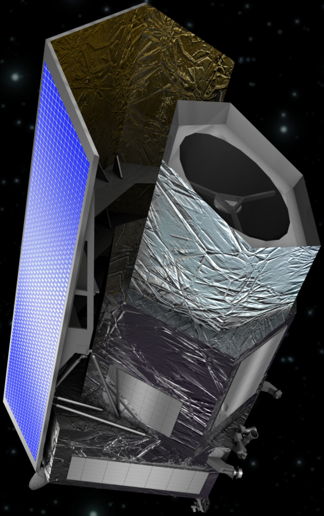

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- Build up an accurate 4D picture of the visible universe
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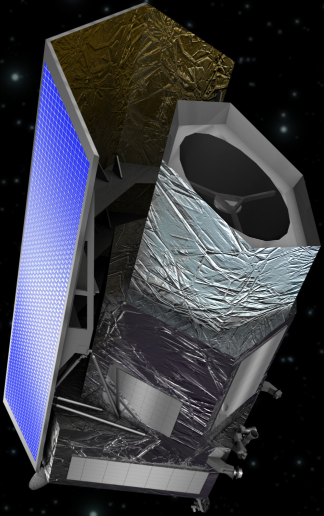
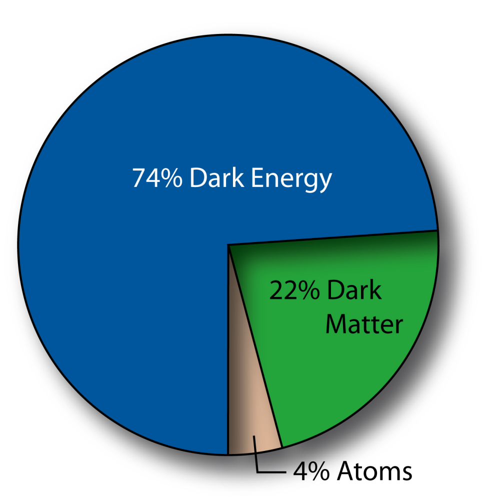
- Will see any variation in dark energy across space and time, enabling us to build a better understanding of it.

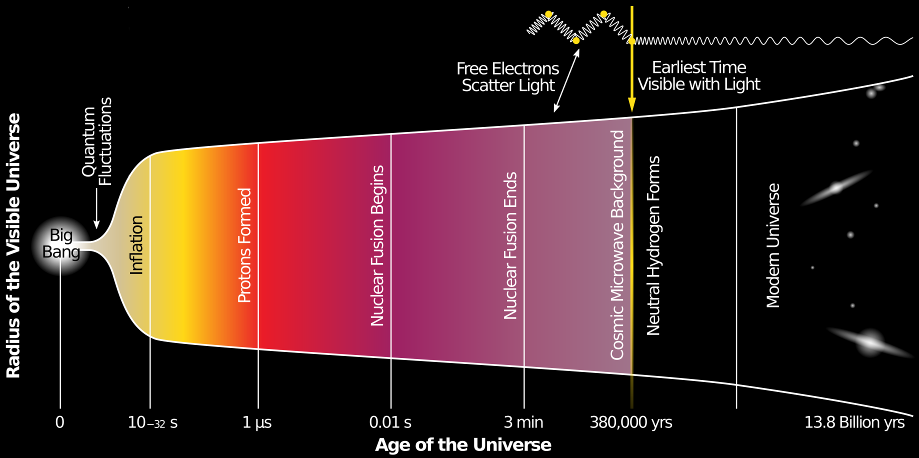


# The Composition of the Universe



# The beginning of time

$t = 0$



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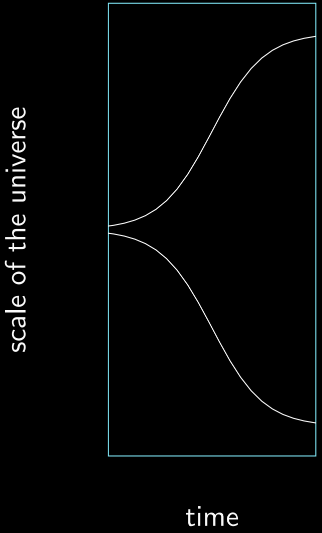

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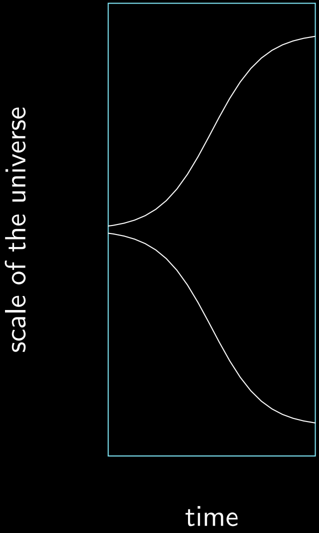
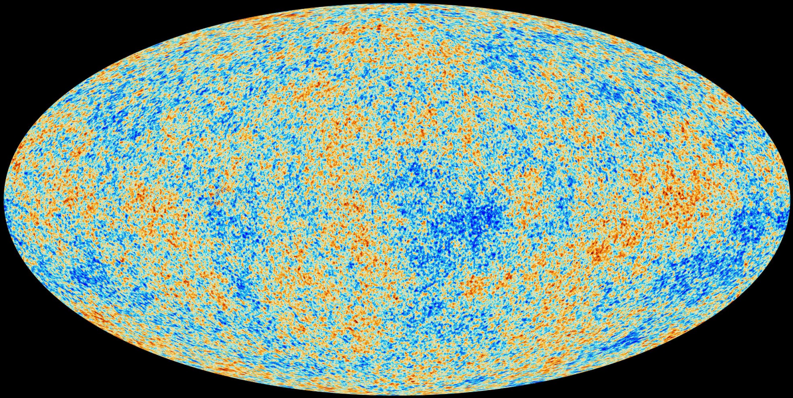
3. We call this accelerated expansion “Inflation”



# Anisotropies in the CMB



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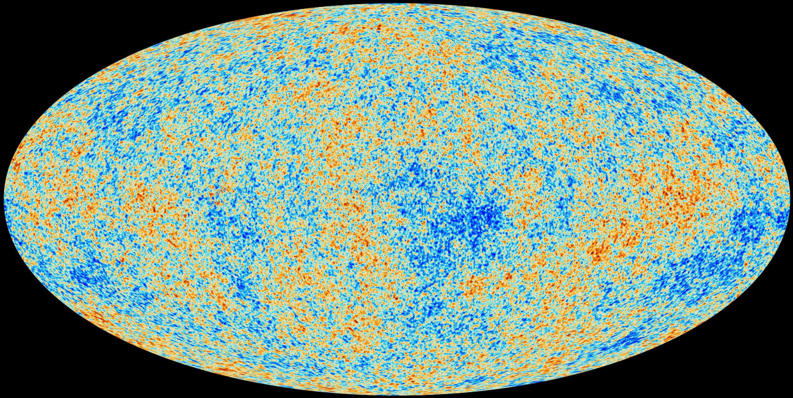

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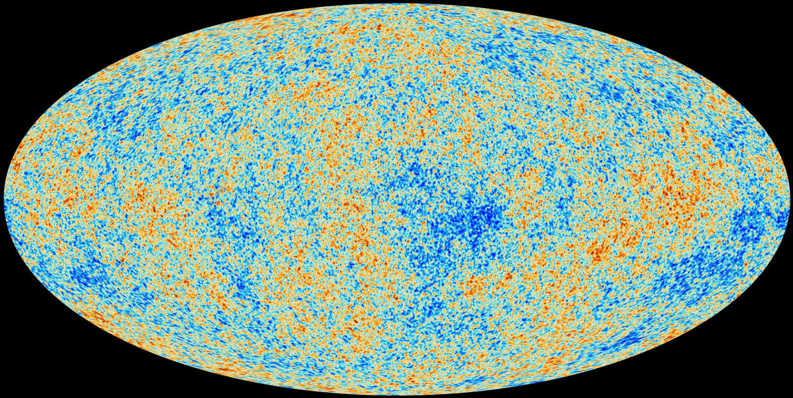

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- The *Theory of Inflation* predicts these inhomogeneities,
- Which are created by *quantum fluctuations*.
- These are the seeds of structure formation in the universe.

# The dawn of a new era in astronomy

**The rise of the mega-telescope**

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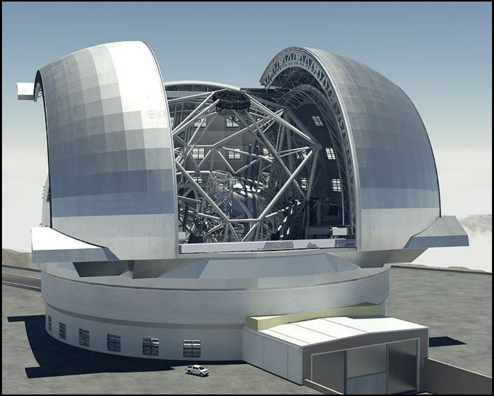
- Imaging Hydrogen through the universe will build a 3D picture of the first ripples of structure.

# European Extremely Large Telescope (E-ELT)

**2024: Ground-based 40m optical telescope**



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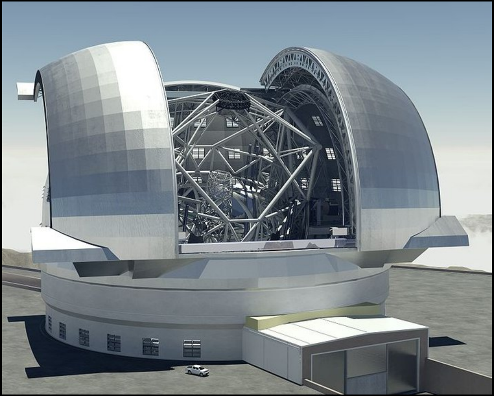

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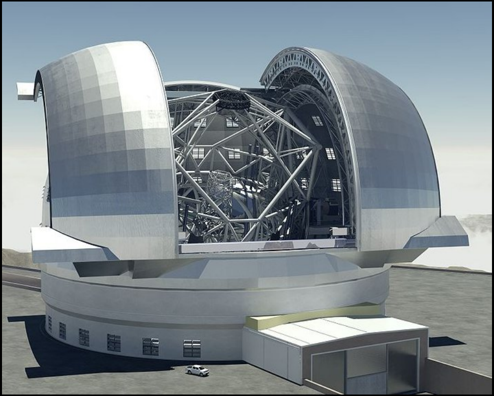

# European Extremely Large Telescope (E-ELT)

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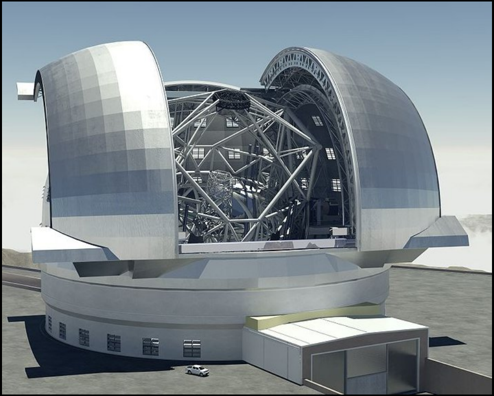
- Search for extrasolar planets.
- Over a 20 year baseline, it will be sufficiently accurate to watch the universe expand.



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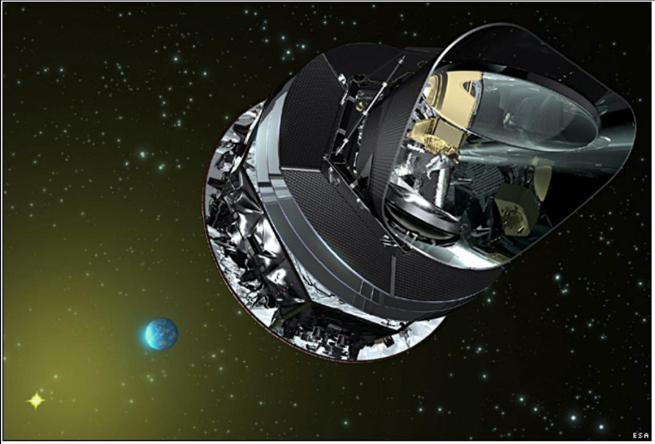

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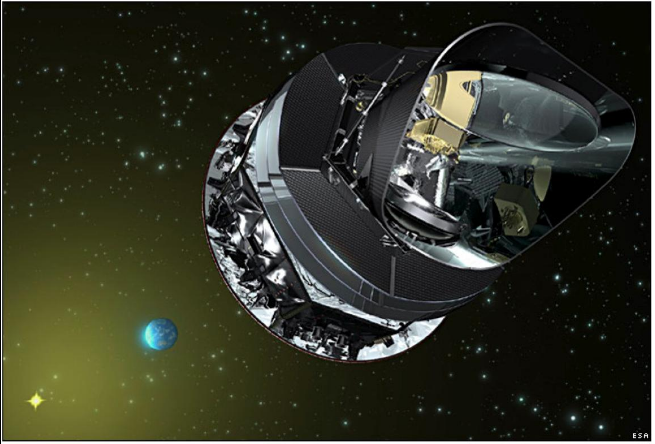


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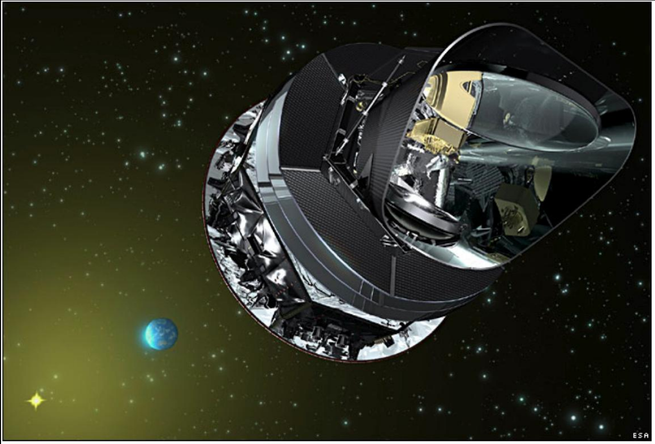


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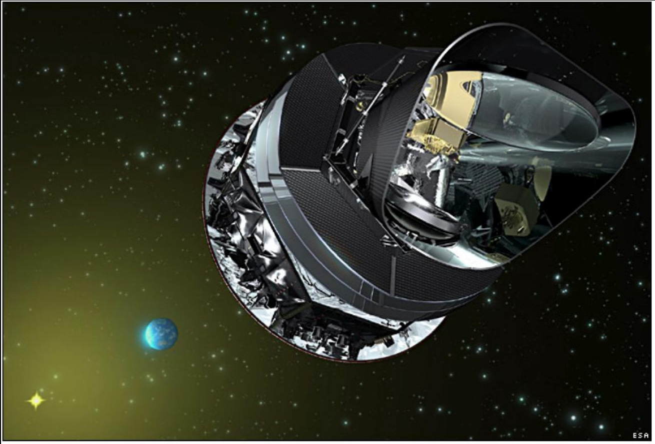
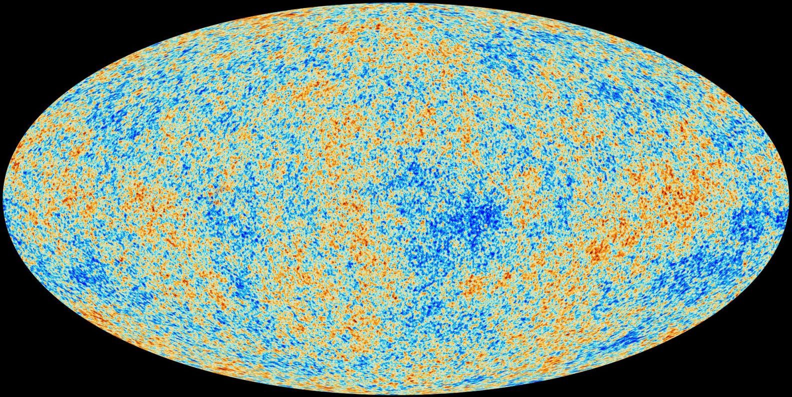



- Next-generation CMB mapping: polarisation
- Cataloguing galaxy clusters
- Providing essential data for accurate exploration of cosmological theories

# Summary



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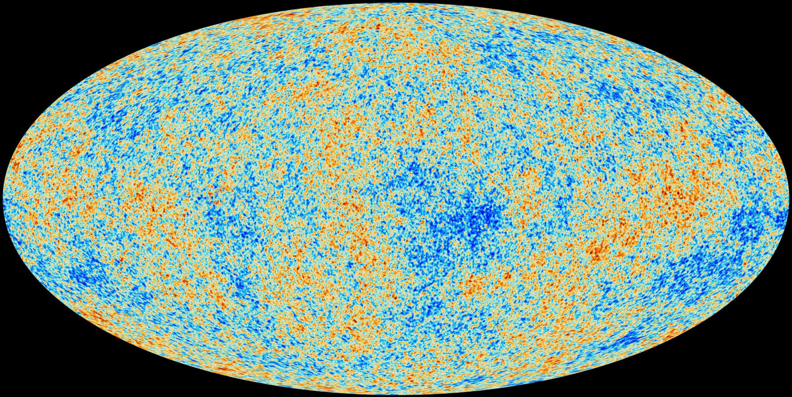

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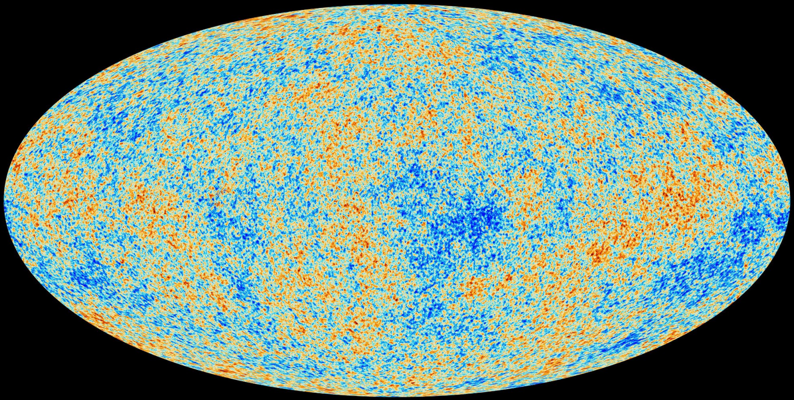

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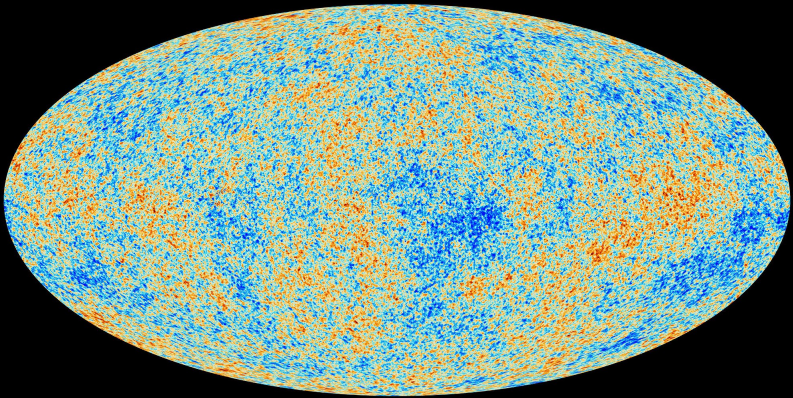

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- There are many unsolved theoretical problems,
- And a wealth of data to explore them with.

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**Cosmology is the study of the universe as a whole**

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